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# PRELIMINARY PALEOMAGNETIC DATA FROM THE JURASSIC HUMBOLDT LOPOLITH, WEST-CENTRAL NEVADA: EVIDENCE FOR THRUST BELT ROTATION IN THE FENCEMAKER ALLOCHTHON

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Abstract. Layered and massive gabbros of the Humboldt Lopolith and extrusive equivalents in west-central Nevada yield well-grouped paleomagnetic data on a site basis. With corrections for attitudes of compositional layering in gabbros and flow contacts in lavas ($k_{corr}/k_{in\ situ}$ = 5.5), the mean of 15 site means is: D = 307°, I = 21°, $\alpha_{95}$ = 9°, k = 17. While K-Ar data imply a Late Jurassic age, we prefer an older, probably Early Jurassic, age for the lopolith and its magnetization based upon magnetization data and relations of these rocks with host strata. Compared with Early Jurassic expected directions, the data reveal a discordancy in declination of -37° (+/- 14-22°) which implies a significant counterclockwise rotation of the lopolith. Local rotation of Fencemaker allochthon units most likely occurred when southeasterly-directed thrust sheets obliquely contacted a generally north-south trending shelf margin. Mesozoic deformation will be more accurately estimated with better age control and measurement of possible Cenozoic crustal rotation.

## Introduction

Layered pre-Tertiary rocks of the central and western Great Basin constitute a series of structurally dismembered lithotectonic assemblages involved in three episodes of regional contractional deformation from Late Devonian to Late Cretaceous time [Speed, 1978a,b, 1979, 1983; Oldow, 1981, 1983, 1984]. Regional stratigraphic relations imply that at least some of the Triassic to Early Cretaceous rocks (basinal rocks overlain by shallow marine and subaerial sedimentary rocks) in these assemblages were close to the continental margin at the time of their deposition, and were latitudinally continuous with correlable units on the continent [Oldow, 1984; Russell, 1984; Lupe and Silberling, mscpt. in prep.]. Primary magnetizations, which could be used to provide paleolatitude and paleo-orientation information for these units with respect to stable North America, may no longer exist as a result of widespread late Mesozoic magmatism and/or penetrative deformation [e.g., Geissman et al., 1984; Russell et al., 1982]. Data from most intrusive rocks may be of limited value because they are late-syntectonic or post-tectonic and the local paleohorizontal is difficult to infer [Beck, 1980; Geissman et al., 1982, 1984].

We have begun a study of gabbros, equivalent extrusive rocks, and associated metasedimentary rocks belonging to, or related to, the Humboldt Lopolith [Speed, 1963, 1976] in west-central Nevada (Figure 1). The lopolith, of pre-thrusting age, intrudes Triassic and possibly earliest Jurassic rocks of the Mesozoic marine province and is part of the Fencemaker allochthon [Speed, 1978b, 1983; Oldow, 1983, 1984]. Gabbros are generally compositionally layered and/or texturally foliated, largely due to primary magmatic processes [Speed, 1963]. Pre-Late Cretaceous compression and Cenozoic extension have resulted in fragmentation, open folding, and tilting of the lopolith. Our hope that the compositional layering of the gabbros represented the paleohorizontal, and therefore that the age of the lopolith magnetization could be evaluated relative to deformation, appears to have been confirmed. A primary magnetization for the lopolith would permit a better understanding of the transport history of the Fencemaker allochthon and the paleolatitude of marine province units at the time of lopolith intrusion.

## Sampling And Methodology

We report data from 15 sites (generally 7 or more samples/site) in compositionally layered gabbros, extrusive equivalents, and clastic rocks within the volcanic sequences, in several structural settings in the West Humboldt, Stillwater, and Clan Alpine Ranges. Samples were usually collected as oriented blocks, as the terrain and access are not favorable to in situ drilling. Natural remanent magnetization data were acquired on a Schonstedt SSM-1a spinner magnetometer, modified with a Hewlett-Packard HEDS 500 count/rotation optical shaft encoder, and interfaced with a Cromemco System Two microprocessor. Progressive alternating field or thermal demagnetization techniques were employed in all cases to adequately resolve all magnetization components. Directions of components were generally determined using vector subtraction. In some cases, great circle ("remagnetization") analysis was required to evaluate whether curvilinear demagnetization behavior reflected systematic overprinting of components which were usually isolated in more straightforward response.

## Results

Examples of common demagnetization behavior by compositionally layered gabbros are shown in Figure 2. Response by other lithologies was generally as straightforward. Site mean directions, before and after simple structural corrections for local tilt of compositional layering or strata which assume horizontal fold axes, are shown in Figure 3. The mean of the corrected site means is: D = 307°, I = 21°, $\alpha_{95}$ = 9°, k = 17, N = 15 sites. We propose that this magnetization is characteristic of the Humboldt Lopolith units, that it predates Cretaceous plutonism, and that

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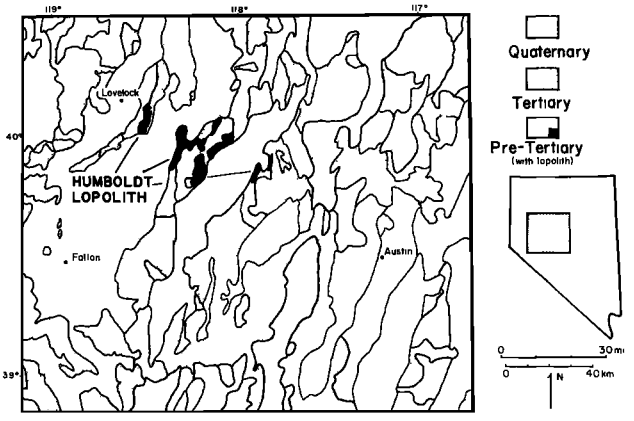



Fig. 1. Location of the Humboldt Lopolith and other pre-Tertiary rocks in west-central Nevada. From east to west, the three ranges containing exposures of the lopolith are the Clan Alpine, Stillwater, and West Humboldt. After Stewart and Carlson [1978].

it most likely reflects the time of initial intrusion and cooling. Our principal observations in support of this statement are as follows. The characteristic magnetization is generally a high corecivity (median destructive fields usually exceed 15 mT), high blocking temperature (but below 580°C) component carried by low-titanium magnetite, as is common for gabbroic intrusions. This component is dual polarity, in some cases at the site level. After structural corrections, an F-ratio test establishes that the improvement in grouping ($k_{corr}/k_{in\ situ} = 5.5$) is significant at the 99% confidence level. The outcome of this exercise is good evidence that compositional layering in several structural regions reflects the paleohorizontal and that the magnetization predates deformation related to Mesozoic thrusting. Had widespread remagnetization occurred later in Mesozoic time, then a north-northwesterly, moderate positive inclination magnetization (and/or antipode) would have existed prior to local Cenozoic tilting. As field evidence clearly indicates that considerable deformation occurred prior to eruption of Oligocene ash-flow tuffs, we have also attempted to restore site directions to a pre-Oligocene setting by correcting, where possible, for components of Cenozoic tilting (Figure 3). In all cases, the site data are not well-grouped about expected Late Jurassic to Late Cretaceous directions.

## Discussion

Interpretation of the Humboldt Lopolith data in terms of major Mesozoic tectonism in west-central Nevada requires an accurate age for the intrusion. Existing K-Ar biotite and hornblende data on lopolith gabbros range from 163 to 144 Myr [Speed and Jones, 1969; Speed and Armstrong, 1971]. We prefer an older, probably Early Jurassic age for the lopolith and suggest that the K-Ar data possibly record major thrusting in Late Jurassic time. In the West Humboldt Range the lopolith intrudes Triassic rocks, placing an older limit of latest Triassic on the age of intrusion (R.C. Speed, personal communication, 1983). In parts of the West

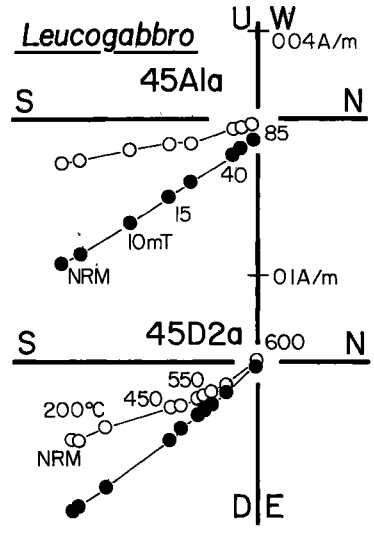



Fig. 2. Orthogonal progressive demagnetization diagrams showing behavior of a fresh, compositionally layered gabbro sample in alternating field and thermal demagnetization. Axes are given in geographic coordinates. Open (closed) symbols refer to vertical (horizontal) projections. Note that at this site primarily reverse polarity magnetizations were recorded.

Humboldt Range, Triassic and earliest Jurassic strata appear to have been deformed to some extent prior to lopolith intrusion. In the Stillwater and Clan Alpine Ranges, the lopolith intrudes the Boyer Ranch Formation. The limiting Early to Late Jurassic age of the Boyer Ranch units is based on the lopolith K-Ar ages [Speed and Jones, 1969]. An older age for the lopolith is not inconsistent with stratigraphic relations.

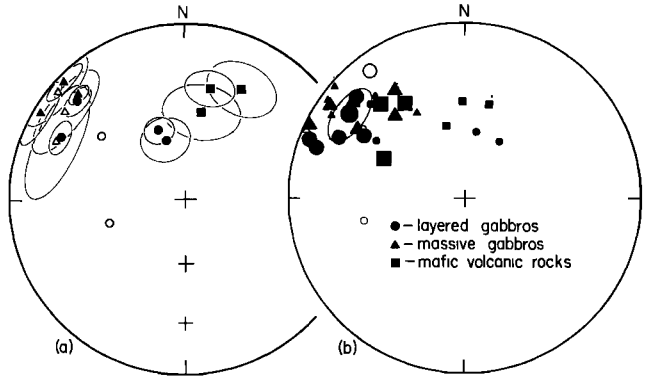



Fig. 3. Equal area stereographic projections of site mean directions for Humboldt Lopolith units. (a) Coordinates refer to in situ orientation; symbols refer to different lithologies. The two site directions without projected cones of 95% confidence were determined using remagnetization circle analysis. (b) site mean directions corrected for attitudes of compositional layering or volcanic/sedimtary strata (large symbols) and local Cenozoic deformation only (small symbols). The mean of the site means is given by a large circle with the projected cone of 95% confidence. Closed (open) symbols refer to lower (upper) hemisphere projections.

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U-Pb dating of lopolith gabbros is in progress (R.C. Speed, personal communication, 1983) and will clarify this interpretation.

In Figure 4 we plot very latest Triassic to earliest Cretaceous expected directions for west-central Nevada and associated rotation and flattening errors and compare these with the Humboldt Lopolith mean. With respect to 190, 200, and 210 Myr expected directions, the lopolith mean is significantly discordant in declination (-37° +/- 14-22°) and insignificantly discordant in inclination (+7° +/- 7°). The declination discordancy implies a considerable counterclockwise rotation of the lopolith and associated units of the Fencemaker allochthon in this area. The rotation is most readily explained as having occurred during active thrusting and near-final emplacement of the allochthon. The trace of the Fencemaker Thrust is irregular; north of the Humboldt Lopolith area a major westward salient of Triassic platformal carbonate rocks exists which is thought to be parautochthonous with respect to the Fencemaker Thrust [Oldow, 1984] (Figure 5). South of the salient, units of the Fencemaker allochthon were rotated counterclockwise as thrust sheets obliquely impinged upon the shelf margin. The presence of the salient, acting as a buttress during eastward to southeastward contraction, may have accentuated counterclockwise rotation in the Humboldt Lopolith area. In Figure 5 we show a series of possible thrust traces (dotted lines) during progressive contraction. A rotation of this nature is similar to that shown by Grubbs and Van der Voo [1976] in the Wyoming-Idaho thrust belt. Ongoing structural studies of allochthonous Triassic and Jurassic sedimentary rocks in the southern Humboldt Range, 10 km north of lopolith exposures, also support considerable counterclockwise rotation of the region (J.S. Oldow, personal communication, 1984).

On a larger scale, the data from the Humboldt Lopolith, providing that they may be compared with Early Jurassic expected directions, preclude paleomagnetically resolvable latitudinal translation of the Mesozoic marine province units of

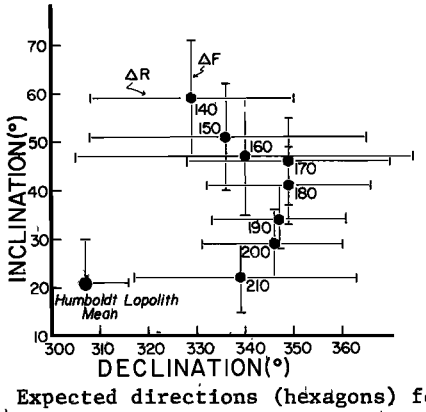



Fig. 4. Expected directions (hexagons) for west-central Nevada between 210 and 140 Myr, calculated from Irving and Irving [1982], and associated rotation and flattening errors(ΔR, ΔF) (bars). The error estimates were determined after Beck [1980] and Demarest [1983], using respective $A_{95}$ values for Irving and Irving's [1982] paleomagnetic poles for each 10 Myr interval and the $\alpha_{95}$ for the Humboldt Lopolith mean.

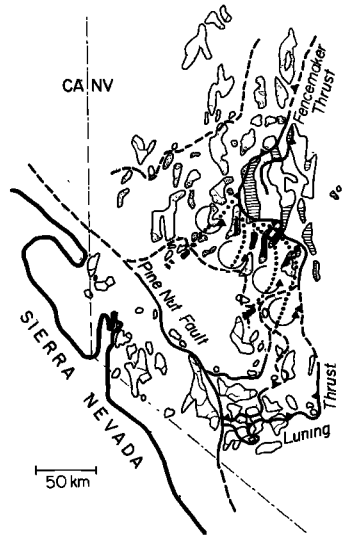



Fig. 5. Generalized map of pre-Tertiary rocks in western Nevada, modified after Oldow [1984]. The Humboldt Lopolith (blackened areas) and other basinal rocks (stippled) form part of a regional dismembered Lovelock Assemblage [Oldow, 1984] thrust southeastward along the Luning-Fencemaker Thrust system. Heavy dotted lines depict a possible sequential development of the Fencemaker Thrust and motion of the allochthon, resulting in counterclockwise rotation of the upper plate during contraction. Horizontally ruled areas represent Mesozoic autochthonous or para-autochthonous assemblages.

western Nevada in Late Triassic and younger time. The data are not inconsistent with kinematic analyses and stratigraphic correlation work by Oldow [1981, 1983, 1984] and Russell [1984] which suggests hundreds of kilometers of southeastward contraction in Jurassic and Cretaceous time for some assemblages in western Nevada. Mesozoic counterclockwise rotation of at least some portions of the Fencemaker allochthon is consistent with a southeastward direction of contraction.

We suspect an additional complicating factor in our present interpretation of the lopolith data. In this area, ranges all trend in a more northeastward direction than they do to the north or to the south. Late Cenozoic block faulting [Anderson et al., 1983] may also have involved substantial clockwise rotation of crustal blocks, possibly caused by shear along a northwest-southeast trending dextral feature with very weak surface expression. The amount of pre-Cenozoic counterclockwise rotation of the Humboldt Lopolith region would have effectively been reduced by such deformation. We are presently evaluating the possibility of Cenozoic clockwise rotation of the ranges with paleomagnetic data from laterally continuous Oligocene ash-flow tuffs.

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